Example 2.45 The present value of a machine is ₹40,000 and its value depreciates each year by 10%. Find the estimated value of the machine in the 6th year.

Solution The value of the machine at present is ₹40,000. Since it is depreciated at the rate of 10% after one year the value of the machine is 90% of the initial value.

That is the value of the machine at the end of the first year is $40,000 \times \frac{90}{100}$

After two years, the value of the machine is 90% of the value in the first year.

Value of the machine at the end of the 2nd year is $40{,}000 \times \left(\frac{90}{100}\right)^2$

Continuing this way, the value of the machine depreciates in the following way as

$$40000,\ 40000 \times \frac{90}{100},\ 40000 \times \left(\frac{90}{100}\right)^2 \ldots$$

This sequence is in the form of G.P. with first term 40,000 and common ratio $\frac{90}{100}$. For finding the value of the machine at the end of 5th year (i.e. in 6th year), we need to find the sixth term of this G.P.

Thus, $n = 6,\ a = 40{,}000,\ r = \frac{90}{100}$.

Using $t_n = ar^{n-1}$, we have $t_6 = 40{,}000 \times \left(\frac{90}{100}\right)^{6-1} = 40000 \times \left(\frac{90}{100}\right)^5$

$$t_6 = 40{,}000 \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10} = 23619.6$$

Therefore the value of the machine in 6th year = ₹23619.60

Exercise 2.7

1. Which of the following sequences are in G.P.?

   (i) 3, 9, 27, 81,… (ii) 4, 44, 444, 4444,... (iii) 0.5, 0.05, 0.005,…

   (iv) $\frac{1}{3}, \frac{1}{6}, \frac{1}{12}, \ldots$ (v) $1, -5, 25, -125, \ldots$ (vi) 120, 60, 30, 18,… (vii) $16, 4, 1, \frac{1}{4}, \ldots$

2. Write the first three terms of the G.P. whose first term and the common ratio are given below.

   (i) $a = 6,\ r = 3$ (ii) $a = \sqrt{2},\ r = \sqrt{2}$ (iii) $a = 1000,\ r = \frac{2}{5}$

3. In a G.P. 729, 243, 81,… find t_7.

4. Find x so that $x+6$, $x+12$ and $x+15$ are consecutive terms of a Geometric Progression.

5. Find the number of terms in the following G.P.

   (i) 4, 8, 16,…, 8192? (ii) $\frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \ldots, \frac{1}{2187}$

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